

Tunable Graphene Frameworks Enable Advanced Phase Change Composites with both Ultrahigh Thermal Conductivity and Latent Heat

Yapeng Chen, Xiyang Su, Meizhen Liao, Jinhong Yu, and Jinglei Yang*

Developing advanced phase change materials (PCMs) with simultaneously enhanced thermal conductivity and preserved large latent heat is critical for efficient thermal management of electronics, enabling rapid heat harvesting within narrow temperature windows. Incorporating prefabricated high-thermal-conductivity frameworks into organic PCM matrices represents the most promising strategy to boost heat transfer while retaining intrinsic energy storage capacity. However, conventional casting methods face fundamental limitations: strong interlayer interactions and gravity-induced densification hinder the construction of controllable nanosheet frameworks. Herein, a solvent-surface-tension-regulated approach is proposed to fabricate hierarchically aligned graphene frameworks. The resulting eicosane-based composite achieves an exceptional in-plane thermal conductivity of $71.6 \text{ W m}^{-1} \text{ K}^{-1}$ at 27.8 vol.% graphene loading while maintaining a high latent heat of 123.8 J g^{-1} . Capillary channels within the oriented framework simultaneously enhance eicosane absorption capability from release. Through infrared thermography, finite element simulations, and performance benchmarking, superior thermal conduction and phase transition kinetics are validated. This work establishes a new paradigm for constructing orientation-controlled thermal-conductive networks, advancing next-generation PCM composites in thermal management for high-power electronics.

buffering, and temperature regulation, especially in modern electronics requiring higher integration, miniaturization, and intelligence. However, a fundamental limitation of all phase change materials (PCMs) is their inherently low thermal conductivity, typically below $0.5 \text{ W m}^{-1} \text{ K}^{-1}$.^[1–3] This deficiency severely impedes their practical application in scenarios demanding rapid thermal harvesting and efficient thermal management. Furthermore, the inherent phase transition from solid to liquid poses additional challenges: it can contaminate devices during heat absorption and restricts PCM use in electronics thermal management.^[4–6] Therefore, developing strategies to simultaneously enhance both the thermal conductivity and leakage-proof capacity of PCMs is essential to unlock their full potential for efficient heat harvesting, storage, and management across diverse fields.

Despite extensive research, achieving phase change composites (PCCs) that simultaneously exhibit high thermal conductivity and large latent heat remains a significant challenge.^[7–10] Various strategies

have been pursued to enhance the thermal conductivity of PCM matrices, including filler functionalization, compression-induced construction, and the use of pre-fabricated frameworks.^[6,11–13] However, these approaches often struggle to concurrently establish efficient thermal transport pathways and robust encapsulation for the molten PCM. Chemical functionalization, for instance, can improve interfacial interactions to some extent but typically fails to create continuous, high-conductivity networks.^[14–16] Compression-induced methods are generally constrained by scale and low filler loading, resulting in composites with limited thermal storage capacity and insufficient thermal conductivity.^[17–19] Pre-fabricated frameworks hold promise, but their effectiveness is frequently hampered by weak inter-filler connections, leading to high interfacial thermal resistance and inadequate leakage prevention during phase transition.^[20,21] Furthermore, a critical bottleneck lies in the difficulty of achieving both high filler loading and controlled alignment of thermally anisotropic nanofillers (e.g., graphene, boron nitride). Conventional alignment techniques, such as freeze-casting, often operate effectively only at low loadings,

1. Introduction

High-performance phase change composites (PCCs) are critically important for thermal energy harvesting, thermal

Y. Chen, X. Su, J. Yang

Department of Mechanical and Aerospace Engineering

Hong Kong University of Science and Technology

Hong Kong 999077, China

E-mail: maeyang@ust.hk

M. Liao, J. Yu

State Key Laboratory of Advanced Marine Materials

Ningbo Institute of Materials Technology and Engineering (NIMTE)

Chinese Academy of Sciences

Ningbo 315201, China

J. Yang

HKUST Shenzhen-Hong Kong Collaborative Innovation Research Institute

Futian, Shenzhen 518000, China

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restricting the formation of dense, interconnected thermal pathways and diminishing the capillarity needed for effective PCM confinement. Consequently, the development of PCCs integrating superior thermal conduction with reliable anti-leakage performance demands novel structural designs that can overcome these intertwined limitations.

In this study, we present a novel strategy for developing PCCs with high latent heat and significantly enhanced thermal conductivity. Our breakthrough approach involves controlling the internal orientation of graphene nanosheets and the gaps between them by precisely regulating the interfacial tension between graphene and the dispersion liquid during vacuum-pressure-assisted assembly. Following infiltration with eicosane, the resulting composite achieved a latent heat of 123.8 J g⁻¹ and a remarkably high thermal conductivity of up to 71.6 W m⁻¹ K⁻¹. These superior properties are attributed to the oriented graphene framework with optimized porous spacing, which completely eliminates leakage after repeated heating cycles. To validate these enhancements, we conducted a series of finite element method (FEM) simulations based on the apparent heat capacity method. Notably, the simulated temperature distributions perfectly matched those captured using an infrared (IR) camera. Furthermore, practical applications—including solar energy harvesting and storage, electronics heat dissipation, and use as a thermal interface material—demonstrated the composite's excellent heat storage and transport capacity, as well as its durability over multiple heating-cooling cycles. Our strategy establishes a new paradigm for developing high-thermal-conductivity PCCs with high latent heat, promising vital roles in energy harvesting, storage, and thermal regulation.

2. Results and Discussion

2.1. Controlled Fabrication and Microstructural Characterization of Aligned Graphene Frameworks

Filtration-casting, a technique capable of constructing continuous frameworks from nanomaterials, is widely employed to fabricate 3D frameworks that enhance the mechanical, electrical, and thermal conductivity of composites.^[22,23] Specifically, the assembly of continuous frameworks using 2D nanosheets—such as graphene, boron nitride nanosheets, MXene, and black phosphorene—has been extensively studied. However, conventional filtration-casting typically results in frameworks where the vast majority of 2D nanosheets are horizontally oriented with minimal interlayer spacing.^[24] This orientation arises from the combined effects of gravity and high surface tension acting on the nanosheets during the casting process.

In contrast, our design achieves continuous graphene frameworks with tunable nanosheet orientation and loading by regulating the surface tension of the dispersion solvent. High-quality graphene nanosheets, confirmed by the Raman spectrum in Figure S1 (Supporting Information), were used for constructing the frameworks. The schematic of our method is presented in Figure 1a, and the measured surface tension of the water/ethanol mixed solvent with varying ethanol content is shown in Figure 1b. Specifically, using a higher water content (lower ethanol ratio), which improves the solvent surface tension, yields a framework with highly ordered graphene nanosheets and improved skeleton

density (Figure 1c,d; Figure S3, Supporting Information). This ordered orientation was further confirmed by polarized Raman spectroscopy. As shown in Figure 1g and Figure S4 (Supporting Information), the normalized G-band peak intensity exhibits a characteristic spindle-shaped angular dependence, indicative of high nanosheet alignment, corroborating the SEM observations in both Figure 1 and Figure S2 (Supporting Information).

Conversely, increasing the ethanol content disrupts this order. At an ethanol volume content of 40%, the graphene nanosheets become randomly oriented within the framework (Figure 1f,g). This random structure is supported by the normalized G-band peak intensity in polarized Raman spectra (Figure 1h), which shows negligible angular dependence. Compared to frameworks produced by conventional filtration-casting, our method enables significantly more uniform and tunable long-range orientation of graphene nanosheets.

These results collectively demonstrate that the microstructure of graphene aerogels can be effectively regulated by controlling the solvent surface tension. Finally, the prepared graphene aerogels were infiltrated with molten eicosane (the phase change matrix) under vacuum to produce phase change composites with enhanced thermal conductivity. FTIR spectra in Figure S5 (Supporting Information) and SEM images in Figure S7 (Supporting Information) reveal the successful fusion of eicosane and the graphene frameworks.

2.2. Fundamental Thermophysical Properties of Graphene/Eicosane Phase-Change Composites

Figure 2 comprehensively summarizes the key thermal properties of the phase change composites, designated AG/E-1 to AG/E-4, where the naming reflects the increasing ethanol volume content (and consequently decreasing solvent surface tension) used during their graphene framework preparation. An overview of their thermal performance is presented in Figure 2a. Notably, AG/E-1, incorporating a graphene loading of 27.8 vol.%, exhibits an exceptional in-plane thermal conductivity $k_{||}$ of 71.6 W m⁻¹ K⁻¹ alongside a high latent heat of 123.8 J g⁻¹. Despite its obvious anisotropic thermal conductivity (Figure S8, Supporting Information), the combination of ultrahigh thermal conductivity and significant latent heat storage represents a major advancement for phase change composites. Furthermore, the aligned graphene framework within AG/E-1 imparts excellent encapsulation capacity for the phase change matrix at elevated temperatures. This is evidenced by the stable DSC curve shown in Figure 2b and the effective leakage-proof performance observed post-heating. Notably, the SEM images of the AG/E composites (Figure S7, Supporting Information) reveal that the skeleton structure is intact, ensuring high thermal conductivity and anti-leakage capabilities.

Thermogravimetric analysis (TGA) conducted under an air atmosphere (Figure 2c) provides two key insights: the composites demonstrate excellent thermal stability below 240 °C, and the TGA curves reveal variations in the final residue mass, indicative of the differing graphene loadings achieved across the AG/E series prepared under controlled solvent conditions. The remarkable efficiency of thermal conductivity enhancement achieved by our filler framework is quantified using the Thermal

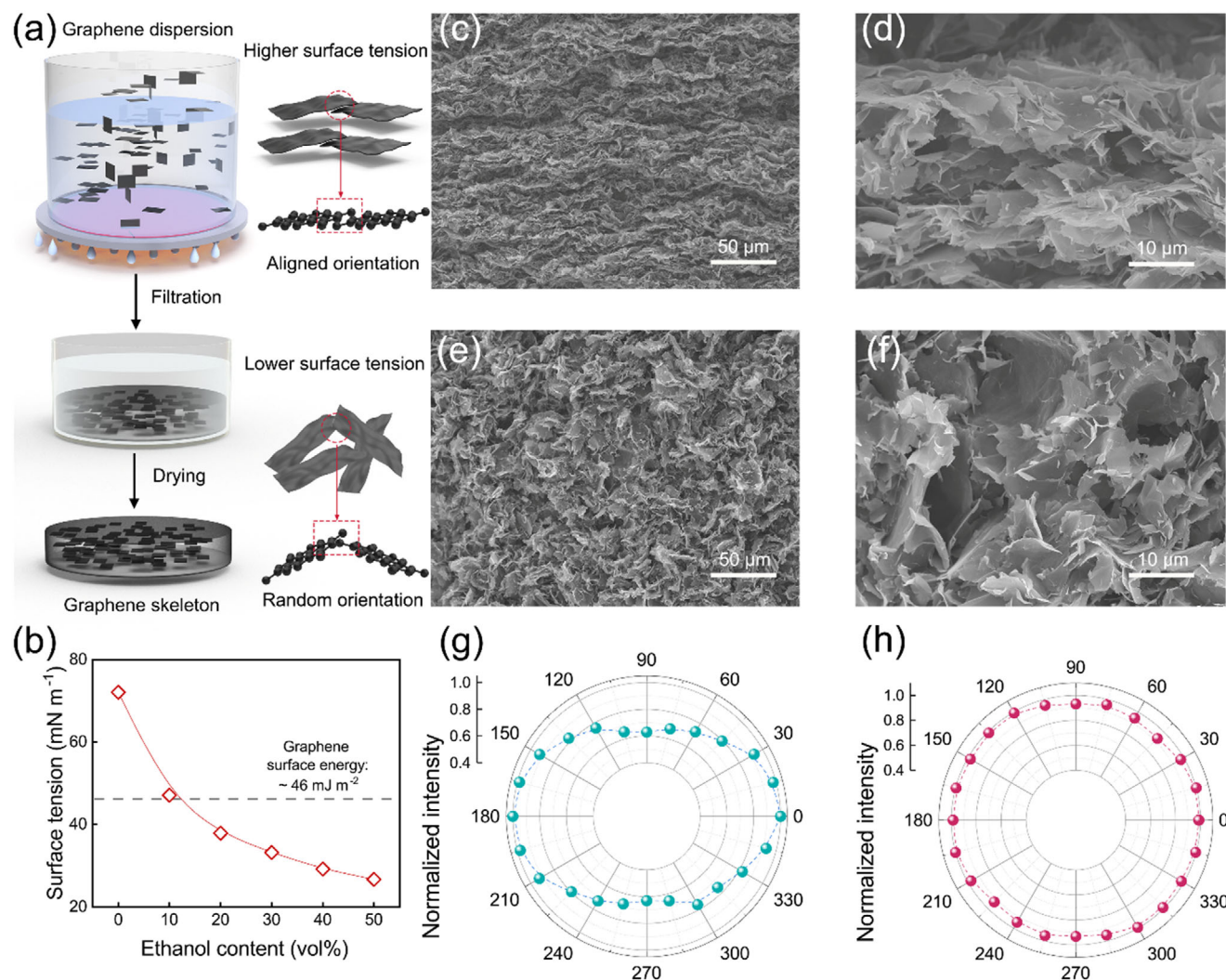


Figure 1. Controllable graphene framework preparation and characterization. a) Schematic illustration of the preparation process for a graphene skeleton with tunable nanosheet orientation. b) Surface tension of the water/ethanol mixed solvent as a function of ethanol volume percentage (vol.%). c) Cross-sectional SEM image and d) higher-magnification view of the graphene aerogel prepared with 10 vol.% ethanol (denoted AG-1), showing aligned nanosheets. e) Cross-sectional SEM image and f) higher-magnification view of the graphene aerogel prepared with 40 vol.% ethanol (denoted AG-4), showing random nanosheet orientation. g) Normalized G-band peak intensity as a function of polarization angle for AG-1 (aligned), exhibiting characteristic anisotropic behavior. h) Normalized G-band peak intensity as a function of polarization angle for AG-4 (random), exhibiting isotropic behavior.

Conductivity Enhancement (TCE) metric (calculation details provided in the Supporting Information). As clearly depicted in Figure 2d, the TCE of our composites, exceeding $\approx 22\,000\%$, surpasses all previously reported values – including those utilizing graphitized carbon frameworks – demonstrating an exceptional level of improvement efficiency.

Beyond thermal conductivity, thermal effusivity (e), defined as:

$$e = \sqrt{k\rho L} \quad (1)$$

where k is thermal conductivity, ρ is density, and L is latent heat, serves as a critical parameter reflecting a material's combined heat storage and rapid transport capability.^[50] Comparative analysis (Figure 2e) confirms that AG/E-1 possesses superior

thermal effusivity among benchmarked phase change composites, positioning it as an outstanding candidate for applications demanding rapid absorption or release of substantial thermal energy.

The impact of enhanced thermal conductivity on thermal response kinetics is evident from DSC measurements. Comparing the curves of pure eicosane and AG/E-1 acquired at heating rates ranging from 10 to $90^\circ\text{C min}^{-1}$ (Figure 2f; Figure S5, Supporting Information), AG/E-1 consistently exhibits lower onset (T_{onset}) and endset (T_{endset}) temperatures. This shift primarily stems from graphene-induced nanoconfinement effects that thermodynamically depress the composite's phase-change temperature. Crucially, while additives drive intrinsic melting point reduction, the high thermal conductivity minimizes thermal lag—sharpening DSC peaks and

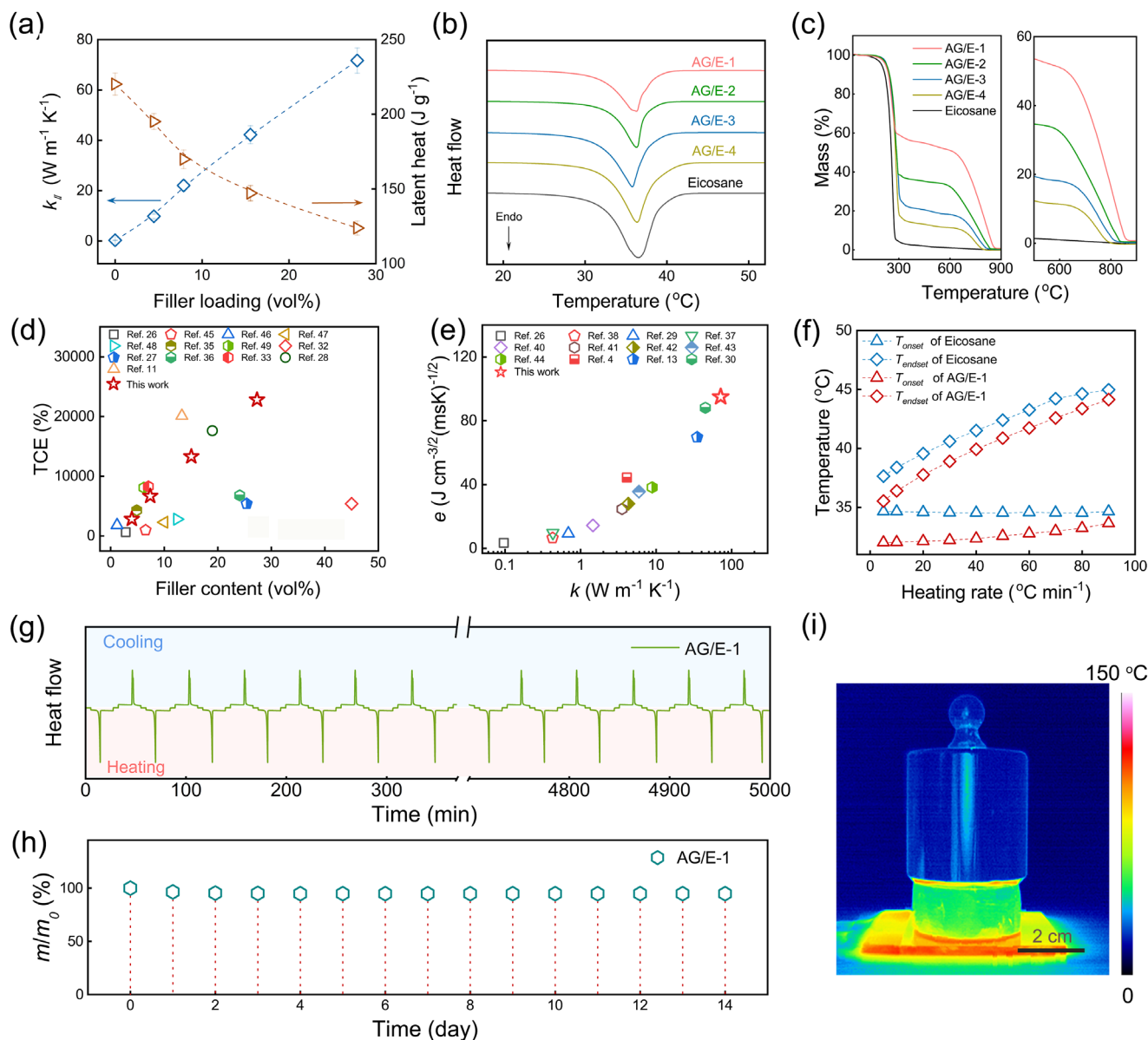


Figure 2. Thermal property characterization of graphene/eicosane phase change composites. a) In-plane thermal conductivity and latent heat of aligned graphene/eicosane composites (AG/E series) as a function of graphene loading. b) DSC curves of AG/E composites with varying graphene loading. c) TGA curves of AG/E composites measured in air. Comparison of (d) thermal conductivity enhancement (TCE) and e) thermal effusivity ($e = \sqrt{k\rho L}$) between AG/E composites and previously reported graphene-enhanced composites.^[4,11,13,25–49] f) Onset (T_{onset}) and endset (T_{endset}) temperatures for pure eicosane and AG/E-1 composite derived from DSC curves at different heating rates. g) DSC curves for cyclic stability testing of AG/E-1 over 100 heating/cooling cycles. h) Weight change of the AG/E-1 under continuous heating under 100 °C. i) Infrared thermal image demonstrating mechanical integrity of AG/E-1 sandwiched between a heat source and a 500 g weight after 20 min of heating.

advancing apparent transition temperatures through rapid heat equilibration.

Long-term operational viability necessitates excellent thermal stability, durability, and mechanical strength. AG/E-1 demonstrated outstanding thermal reliability through rigorous testing: subjected to 100 consecutive heating/cooling cycles (Figure 2g), it exhibited minimal latent heat fluctuation ($\approx 5\%$ decrease, Figure S11, Supporting Information), confirming exceptional stability and durability (Figure S10, Supporting Information). Furthermore, leakage resistance characterization under continu-

ous 100 °C exposure revealed exceptional containment capabilities, with $<6\%$ total mass loss after 300 h of thermal stress (Figure 2h). Mechanical robustness was systematically validated through skeleton/composite compression tests (Figure S12, Supporting Information) and thermomechanical analysis (TMA, Figure S13, Supporting Information). An IR image captured after heating AG/E-1 for 20 min while supporting a 500 g weight (Figure 2i) shows uniform temperature distribution and no structural compromise, unequivocally confirming the composite's exceptional mechanical integrity.

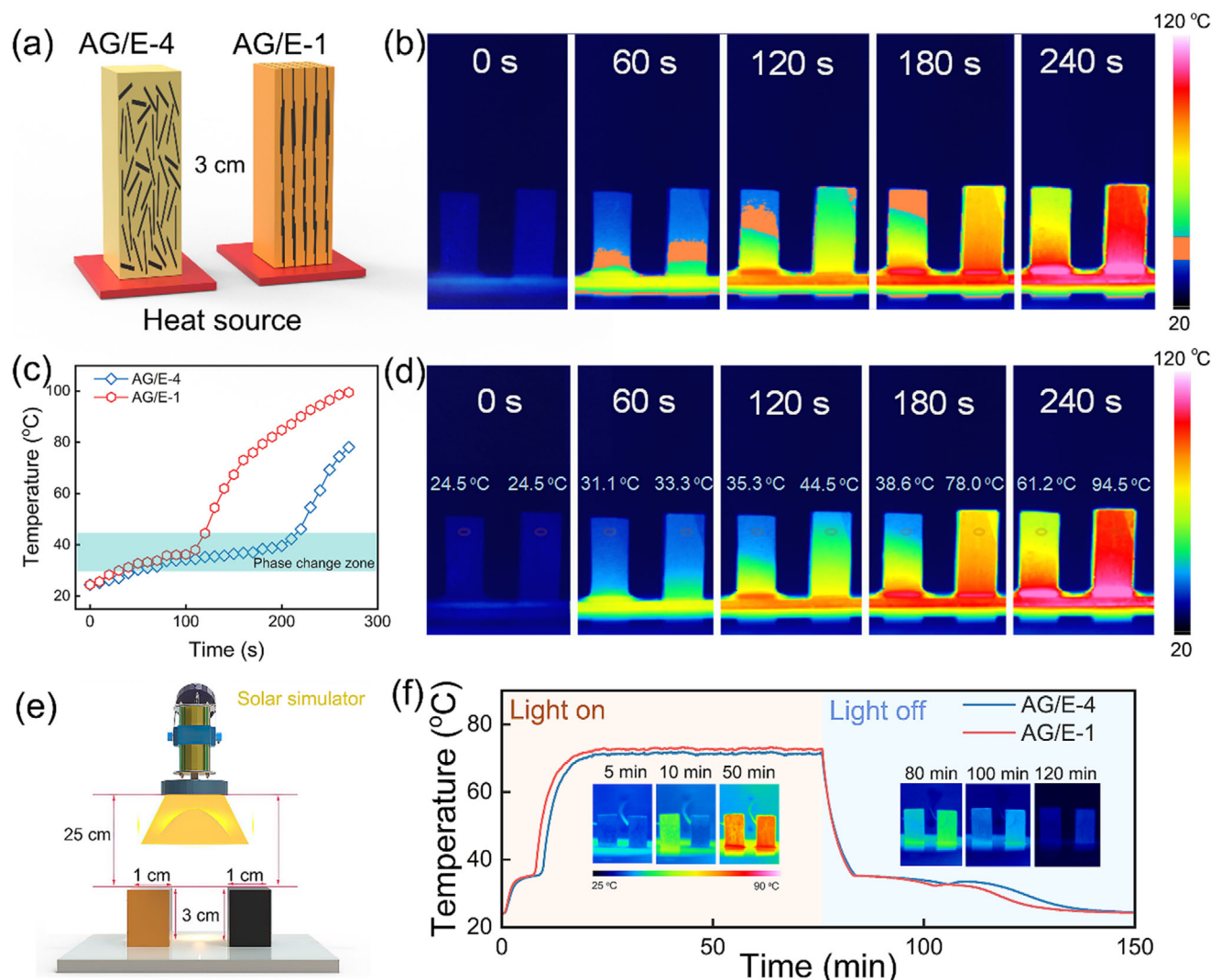


Figure 3. Visualized thermal conductivity, thermal storage performance characterization, and photothermal performance characterization. a) Test system configuration for demonstrating the transient heat transfer capacity of the phase change composite. Infrared images b, c) corresponding to the temperature showing the temperature evolution of the samples. d) Comparison of heat transfer properties along the preferred direction based on the FEM simulated results. e) Schematic diagram of solar-thermal energy conversion characterization system. f) IR camera images of aligned graphene/eicosane composites during the solar charging process and time-temperature evolution curves.

2.3. Experimental Investigation of Dynamic Thermal Transport and Phase-Change Kinetics

To visually demonstrate the enhanced thermal conductivity and highlight the impact of graphene orientation, we recorded the temperature distribution and rise in cuboid-shaped AG/E-1 and AG/E-4 samples ($10 \times 10 \times 30 \text{ mm}^3$) using an IR camera while heating them on a ceramic heater (Figure 3a schematic). Both samples were coated with a thin graphite layer to ensure identical infrared emittance. Figure 3b captures the phase transition evolution. The orange band in the color bar indicates the phase change temperature range (30–40 °C). Both composites initiated solid-to-liquid transition before 60 s; however, AG/E-1 exhibited significantly faster phase change kinetics than AG/E-4. By 120 s, AG/E-1 had nearly completed its phase transition, while AG/E-4 was only halfway

through. This acceleration directly results from AG/E-1's superior thermal conductivity, enabling rapid heat distribution throughout the material. Further confirmation comes from temperature profiles (Figure 3c). AG/E-4 displayed slower heating rates and prolonged phase transition times due to its lower thermal conductivity (despite higher latent heat), contrasting sharply with AG/E-1's performance. After 240 s of heating (Figure 3d), AG/E-1 achieved substantially higher temperatures than AG/E-4, demonstrating that the oriented graphene nanosheets dramatically enhance transient thermal diffusion efficiency.

Beyond thermal management, AG/E composites show exceptional promise for solar-thermal energy conversion, transforming absorbed sunlight into stored thermal energy via PCM phase transitions. Figure 3e illustrates the test setup: a xenon lamp (1.5 sun intensity) irradiated identically sized AG/E-1 and AG/E-4

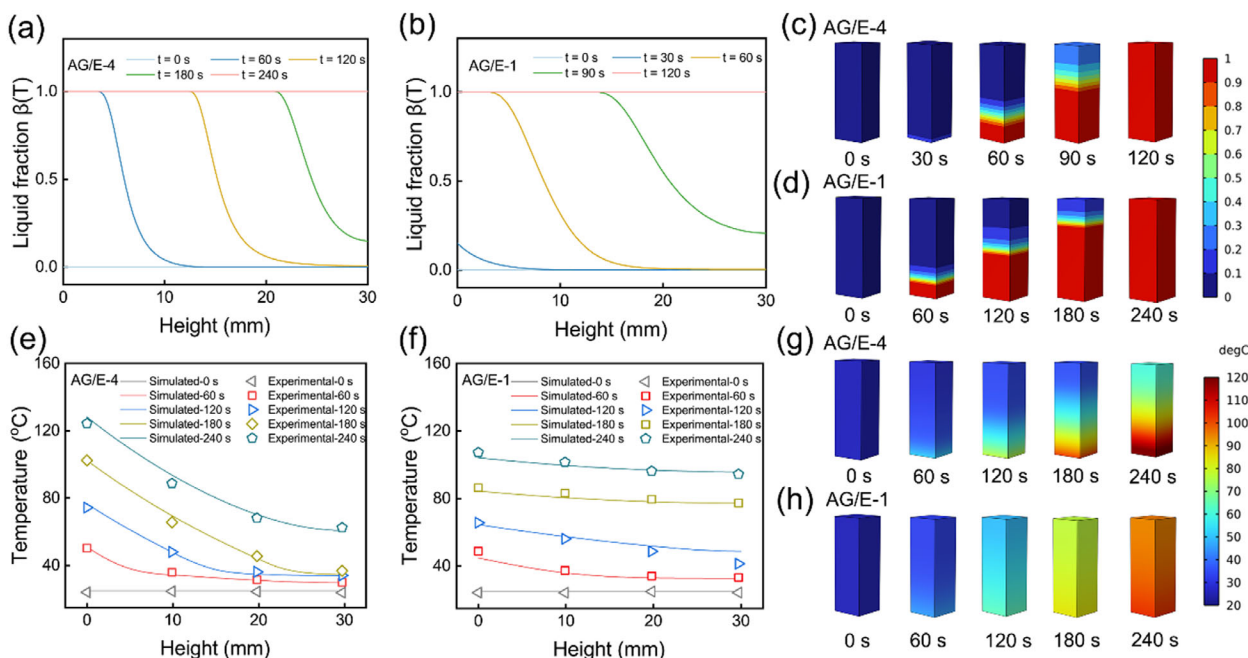


Figure 4. Finite element analysis of phase transition dynamics and thermal gradients in PCM composites. Temporal evolution of liquid fraction β profiles at varying heights for AG/E-4 (a) and AG/E-1 (b), quantitatively characterizing phase change progression. c, d) Spatial distribution of liquid fraction during phase transition for AG/E-4 (c) and AG/E-1 (d), illustrating distinct transformation patterns. e, f) Comparative analysis of simulated and experimentally measured temperature profiles at different elevations for AG/E-4 (e) and AG/E-1 (f), demonstrating excellent model-experiment agreement. g, h) Thermographic visualization of temperature field development in AG/E-4 (g) and AG/E-1 (h), highlighting contrasting heat diffusion characteristics. The size of the models is both $10 \times 10 \times 30 \text{ mm}^3$.

samples (diameter: 1 cm, height: 3 cm) in quartz crucibles. Four thermocouples monitored real-time temperatures in the middle, supplemented by IR imaging.

Key results (Figure 3f) show both composites heating over time under solar simulation. Crucially, during steady-state charging (10–80 min), AG/E-1 maintained a minimal thermal gradient (top: 71.7°C , bottom: 73.1°C), indicating efficient heat distribution. Its extended temperature plateau during cooling confirms effective solar energy harvesting and thermal storage. AG/E-4 also demonstrated high photothermal conversion efficiency relative to benchmark materials.

2.4. Modeling of Phase-Transition Dynamics and Thermal Gradient Evolution

To gain deeper insights into the phase transition behavior and thermal conduction processes of our AG/E composites—which exhibit enhanced thermal conductivity and high latent heat due to orientation-controlled graphene frameworks—we performed finite element method (FEM) simulations. These simulations employed the Apparent Heat Capacity Method with the model ($10 \times 10 \times 30 \text{ mm}^3$) in Figure S14 (Supporting Information) and parameters in Table S1 (Supporting Information), where two critical parameters govern the phase change dynamics:

1. Latent heat L (energy absorbed/released per unit mass during complete phase transition)

2. β defined as:

$$\frac{1}{2} \frac{\theta_L - \theta_S}{\theta_L + \theta_S} \quad (2)$$

where $\theta_L(0-1)$ and $\theta_S(0-1)$ are the phase indicators of liquid and solid fraction, respectively.

The governing transient heat conduction equation is:

$$\rho \left(C_{app} + L \frac{\partial \beta}{\partial T} \right) \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (3)$$

where ρ = density (held constant, independent of temperature/phase change), k = thermal conductivity, Q = internal heat source, $\frac{\partial T}{\partial t}$ = temperature time-derivative.

To eliminate interfacial thermal resistance effects between the ceramic heater and composite, our simulation implemented a temperature-driven boundary condition at the AG/E bottom surface. This temperature profile ($T_{bottom} = 24^\circ\text{C} + 0.36t$ for AG/E-1 and $T_{bottom} = 24^\circ\text{C} + 0.42t$ for the AG/E-4, with t in seconds) were directly imported from experimental IR camera data (Figure S15, Supporting Information). Model parameters and implementation details are provided in the Supporting Information.

The simulations yielded key insights validated against experimental results. Analysis of phase transition progression revealed that after 120 s of heating, AG/E-4 exhibited complete melting only up to 13 mm height, with merely initial melting observed above 20 mm (Figure 4a; Figure S16, Supporting

Information). In contrast, AG/E-1—benefiting from its superior thermal conductivity—achieved full phase transition throughout the entire sample within the same timeframe (Figure 4b; Figure S16, Supporting Information). This differential behavior was further visualized through the evolution of liquid fraction β (Figure 4c,d). Temperature profiles across various heights (0, 10, 20, 30 mm) and time points (0–240 s) showed perfect agreement between FEM results and experimental IR measurements for both composites, robustly validating both the simulation model and the measured thermal conductivity enhancement.

Further thermal gradient analysis revealed significantly steeper temperature distributions in AG/E-4 compared to AG/E-1, directly attributable to its reduced thermal conductivity (Figure 4e,f; Figure S17, Supporting Information). Experimentally, this manifested as pronounced heat accumulation and elevated temperatures at the base of AG/E-4, while its upper region exhibited substantial thermal lag—findings fully consistent with experimental thermal imaging data (Figure 4g,h). Additionally, since pure eicosane melts during heating, we simulated its post-heating temperature distribution (Figure S18, Supporting Information) and phase-change evolution (Figure S19, Supporting Information) under identical heating conditions to AG/E-1. Collectively, these validated simulations demonstrate that orientation-engineered graphene frameworks fundamentally optimize thermal diffusion pathways. This engineered architecture enables rapid, uniform heat propagation and efficient phase-transition kinetics—critical attributes for high-performance applications, including thermal management in high-power electronics and latent heat energy storage systems.

2.5. Application Performance in Thermal Management

To evaluate the practical potential of our phase change composite for thermal management, we assessed its performance as a thermal interface material using a custom-built apparatus (Figure 5a). The test configuration sandwiched the sample between two components: a cold plate maintained at 20 °C via circulating water and a heat source (ceramic heater). Comparative results (Figure 5b) demonstrate that at increasing heater power densities, the heater temperature (T_{heater}) was consistently lower when using AG/E-1 as the TIM compared to having no TIM. The maximum temperature difference reached 25.9 °C at 15 W cm⁻² (AG/E-1: 50.1 °C vs No TIM: 76.0 °C).

From the linear relationship between the steady-state T_{heater} and power density (Figure 5c), we calculated the equivalent heat transfer coefficients (inverse of slope) for each system: 0.498 °C cm² W⁻¹ for AG/E-1 versus 0.080 °C cm² W⁻¹ for no TIM. This represents an 80% enhancement in cooling efficiency achieved by incorporating AG/E-1. Furthermore, the integrated cooling system underwent a rigorous cyclic thermal shock test involving 3000 on/off cycles at 15 W cm⁻². Figure 5d shows that T_{heater} remained remarkably stable throughout these continuous heating/cooling cycles, highlighting the exceptional thermal conductivity stability of AG/E-1 under dynamic conditions. Combining superior heat transfer capability (80% efficiency gain) with excellent thermal shock resistance demonstrated over 3000 cycles,

AG/E-1 emerges as a highly promising TIM candidate for demanding electronic cooling applications.

To validate the cooling capability of AG/E-1 under real-world operating conditions, we applied it as a thermal interface material to a 30 W high-power LED lamp (Figure 5e). The LED and TIM were secured to the heat sink using four fastening screws. The IR images captured at key points are presented in Figure 5f. Upon reaching steady-state after 120 s of operation, the LED surface temperature was 67.8 °C with AG/E-1, compared to 85.6 °C without a TIM, as shown in the surface temperature evolution of the LED lamp over time in Figure 5g. This represents a significant temperature reduction of 17.8 °C (85.6–67.8 °C = 17.8 °C) attributable to AG/E-1, demonstrating its superior cooling performance compared to the no-TIM baseline and highlighting its potential advantage over conventional thermal pads.

3. Conclusion

In summary, we have developed a solvent interface tension-regulated strategy to fabricate hierarchically aligned graphene frameworks, achieving a breakthrough in phase change composites by simultaneously delivering ultrahigh in-plane thermal conductivity (71.6 W m⁻¹ K⁻¹ at 27.8 vol% loading) and exceptional latent heat retention (123.8 J g⁻¹). This approach leverages orientation-engineered networks for efficient phonon transport while integrating capillary channels that provide superior eicosane encapsulation. Through infrared thermography, performance benchmarking, and finite element simulations, we validated the composite's dual capabilities: rapid heat dissipation and stable energy storage. The demonstrated synergy between these properties establishes a new paradigm for constructing controllable thermal-conductive architectures, positioning our composites as high-performance solutions for next-generation thermal management in power electronics.

4. Experimental Section

Materials: Two primary solvents were used: deionized (DI) water and ethanol. DI water was produced in-house with an electrical resistivity of 18 MΩ·cm. Ethanol (analytical grade) was purchased from Sinopharm Group Chemical Reagent Co., Ltd. Graphene slurry (solid content: 5.0 ± 0.5 wt.%) was obtained from Ningbo Morsh Tech. Co., Ltd. The phase change matrix, eicosane, was purchased from Sinopharm Group Chemical Reagent Co., Ltd. and used as received.

Preparation of Graphene Phase Change Composites: Graphene dispersions were prepared by dispersing the graphene slurry in a mixed solvent of DI water and ethanol under strong magnetic stirring and ultrasonication, achieving a final graphene concentration of 10 mg mL⁻¹. The volume ratio of DI water to ethanol was systematically varied from 9:1 to 6:4. The surface tension of these water/ethanol mixtures, a key parameter influencing graphene assembly, was measured using the capillary rise method at ambient temperature. Self-assembled bulk graphene composites were obtained by vacuum filtration of the dispersions. The resulting wet composites were then dried in an oven at 80 °C to remove the solvent and form porous graphene architectures. Notably, the ethanol ratio in the solvent mixture served as a key parameter to regulate both the internal pore size and the orientation of the graphene nanosheets within the framework. Specifically, higher ethanol ratios resulted in larger pore sizes and poorer graphene nanosheet alignment.

Finally, the phase change composite was fabricated by infiltrating molten eicosane into the dried porous graphene architecture under vacuum conditions.

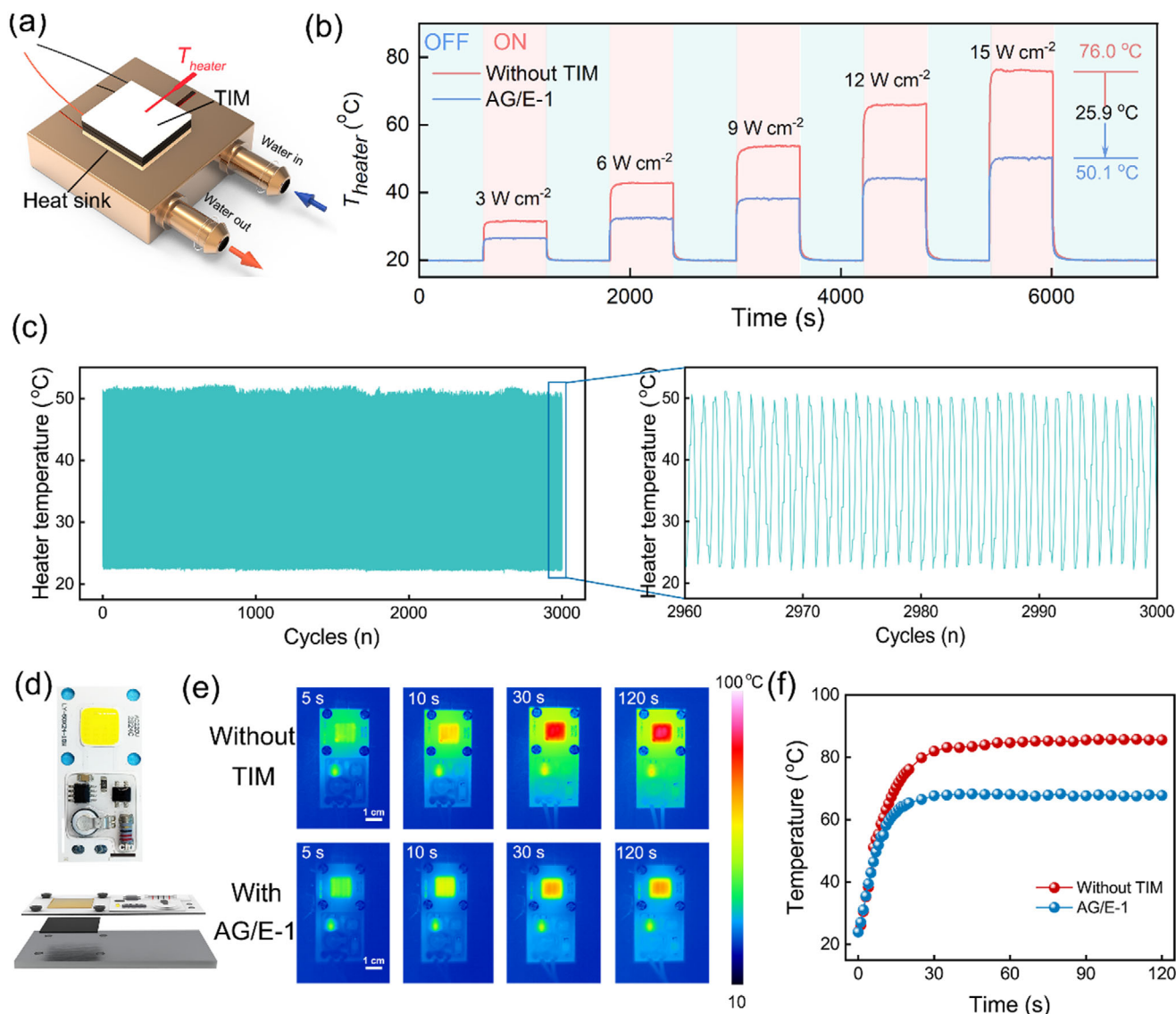


Figure 5. Thermal management performance of phase change composites as thermal interface materials (TIMs). a) Schematic illustration of the experimental setup, where the TIM sample is sandwiched between a temperature-controlled heat sink and a ceramic heater, with a thermocouple monitoring the heater surface temperature. b) Real-time temperature evolution profiles as a function of applied power density (0–15 W cm^{−2}). c) Thermal cycling stability test of the AG/E-1 under repeated heating (15 W cm^{−2})/cooling cycles, demonstrating exceptional thermal shock resistance over 3000 cycles. d) photographic image and schematic diagram of the commercial high-power LED module test configuration. e) IR images captured at (i) t = 0 s, (ii) t = 60 s, and (iii) t = 120 s. f) Time-dependent surface temperature profiles comparing LED operation with (red curve) and without (blue curve) the TIM, demonstrating a significant temperature reduction of $\Delta T = 17.8^\circ\text{C}$ at steady-state operation.

Characterizations: The morphology and alignment of the graphene frameworks were characterized using scanning electron microscopy (SEM, Regulus8230, HITACHI, Japan) at an accelerated voltage of 10 kV. Fourier transform infrared spectroscopy (FT-IR, Nicolet IS50, Thermo Fisher Scientific, USA) was employed to record transmission spectra over the wavenumber range of 400–4000 cm^{−1}, confirming the components of the eicosane-based composites. The surface tension (γ) of the DI water/ethanol mixed solvents was determined using the capillary rise method. The height (Δh) of liquid rise in calibrated glass capillaries of known radius (r) was measured. Surface tension was calculated using the equation:

$$\gamma = \frac{\rho g \Delta h r}{2 \cos \theta} \quad (4)$$

where ρ is the liquid density, g is the gravitational acceleration (9.80 m s^{−2}), and θ is the contact angle (assumed to be 0° for wetting liquids in clean glass capillaries). Measurements were performed in triplicate for each solvent ratio. The latent heat (phase change enthalpy) of the composites was measured using differential scanning calorimetry (DSC 2500, TA Instruments, USA). Samples were analyzed between 0 and 100 °C at various heating and cooling rates. Thermal conductivity (k) was calculated using the formula:

$$k = \alpha \times \rho \times C_p \quad (5)$$

where thermal diffusivity (α) was measured using a laser flash analyzer (LFA 467 Hyper Flash, NETZSCH, Germany), density (ρ) was determined

by mass and volume measurement, and specific heat capacity (C_p) was obtained from the DSC 2500 (TA Instruments, USA). Temperature distributions during heating were monitored using an infrared thermal camera (Fluke Ti480 Pro, USA).

Statistical Analysis: Statistical analyses were performed using Microsoft Excel and OriginPro Learning Edition. The thermal conductivity and latent heat data are presented as mean $\pm$ standard deviation (SD), derived from three independent samples. The dimensions of each sample used for thermal conductivity testing were 10 mm $\times$ 10 mm $\times$ 1 mm. Other experimental details and data analysis are shown in the Experimental Section.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

enhanced thermal conductivity, filtration casting, graphene, phase change composites, phase change kinetics

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